

Week 1

Week 2

Week 3

Week 4

Final Outcome

Week 1 Foundations & Mental Models

Milestone: Understand how LLMs think, master prompt anatomy, run 20+ test prompts confidently

Daily tasks

DAY 1-2

Learn what tokens, temperature, and context windows mean. Read Claude's model card.

DAY 3

Study prompt anatomy: instruction, context, input data, output format. Write 5 prompts dissecting each part.

DAY 4

Practice zero-shot vs few-shot prompting. Compare outputs side-by-side.

DAY 5

Explore role-based prompting. Assign 3 different personas and note output shifts.

DAY 6

Learn about hallucinations and how grounding prompts reduces them.

DAY 7

Review & reflect. Document your top 3 insights. Share a learning on LinkedIn.

Week 1 project

📖 Prompt Anatomy Notebook

Create a structured doc with 10 prompts — each labeled with its anatomy (instruction, context, format, constraints). Include before/after comparisons showing how adding context changed the output.

Beginner

Week 2 Advanced Techniques

Milestone: Use chain-of-thought, XML structuring, and multi-step prompts reliably

Daily tasks

DAY 8-9

Deep-dive into chain-of-thought (CoT) prompting. Write "let's think step by step" prompts and compare to direct prompts.

DAY 10

Learn XML tagging in Claude. Practice using `<context>`, `<instructions>`, `<output>` tags.

DAY 11

Prompt chaining: use output of one prompt as input to the next. Build a 3-step chain.

DAY 12

Practice negative constraints (tell the model what NOT to do). Compare outputs.

DAY 13

Try self-critique prompts — ask Claude to evaluate its own output, then revise.

DAY 14

Review & document all Week 2 techniques. Pick your top 3 favorite and explain why.

Week 2 project

📌 3-Step Prompt Chain

Build a 3-prompt chain for a real task — e.g. "Research → Draft → Polish a LinkedIn post". Document each step, the prompt used, and how the output improved.

Intermediate

Week 3 Real-World Applications

Milestone: Apply prompt engineering to 3+ real domains (content, code, data, education)

Daily tasks

DAY 15-16

Prompt engineering for content creation — marketing copy, blog posts, social captions. Try tone modifiers.

DAY 17

Prompting for code: write prompts to generate, explain, debug, and refactor code snippets.

DAY 18

Data extraction prompts — give Claude a block of unstructured text, extract structured JSON.

DAY 19

Education prompting — design a teaching prompt that explains a concept at 3 difficulty levels.

DAY 20

Evaluation prompts — design a rubric inside a prompt to score and rank outputs.

DAY 21

Review & compile a personal "Prompt Playbook" — your go-to patterns per domain.

Week 3 project

📖 Domain Prompt Playbook

Create a shareable Notion page or PDF with 5 ready-to-use prompt templates across different domains. Each template should include a "when to use" note and a real sample output.

Portfolio-ready

Week 4 Systems, APIs & Mastery

Milestone: Build a system prompt, call the API, and complete a capstone project

Daily tasks

DAY 22-23

Learn system prompts — how they set persistent behavior. Build a reusable system prompt for a use case you care about.

DAY 24

Call the Anthropic API using Python or JS. Send your first programmatic prompt and parse the response.

DAY 25

Experiment with parameters: temperature, max_tokens, top_p. Document how each shifts the output style.

DAY 26

Build a simple prompt testing script — loop through 3 prompt variants and compare outputs.

DAY 27-28

Work on capstone project. Combine system prompt + chaining + domain application into one real tool.

DAY 29-30

Document everything. Write a LinkedIn post or internal post sharing your 30-day journey.

Capstone project

🔗 AI-Powered Mini Tool

Build one real tool using what you've learned — examples: a LinkedIn post generator, a fresher onboarding quiz bot, a product description writer for AalooBites, or a prompt evaluator. Use a system prompt, at least one chain, and an API call.

Capstone

Day 30 What you'll have by the end



Deep mental models

You understand how LLMs process, hallucinate, and respond — not just what buttons to press



Personal playbook

A reusable library of 20+ tested prompts across domains, ready to share or deploy



Live mini-tool

A real API-powered tool or bot you built yourself, end to end



LinkedIn presence

Weekly posts documenting learnings — builds your reputation as an AI practitioner



Training-ready

All content is structured to teach — reuse directly for fresher onboarding sessions



API fluency

Comfortable calling Claude via API, tuning parameters, and building on top of it

30

days

60+

hours learning

8

projects

3

certifications

W1

Week 1 — AI & ML foundations

Days 1–7 · Core concepts, math intuition, first models



W2

Week 2 — Deep learning & neural networks

Days 8–14 · PyTorch basics, CNNs, NLP intro



W3

Week 3 — LLMs, prompt engineering & AI APIs

Days 15–21 · GPT APIs, RAG, LangChain, vector databases



W4

Week 4 — MLOps, deployment & capstone

Days 22–30 · Deploy, monitor, certify, showcase



Recommended free certifications to pursue

- ✓ **Google ML Crash Course certificate** — free, instant, add to LinkedIn. Complete by end of week 1.
- ✓ **DeepLearning.AI short course certificates** — multiple free certs (Prompt Engineering, LangChain, RAG). Stack 3–4 of them.
- ✓ **Kaggle course certificates** — free, respected in the ML community. Complete Python, ML Intro, and NLP tracks.

What you can show at day 30

- ✓ A live, deployed AI application — shareable URL you can send to any recruiter
- ✓ GitHub portfolio with 5–8 repos, clean READMEs, and commit history showing 30 days of work
- ✓ 3–4 certificates on LinkedIn from Google, DeepLearning.AI, and Kaggle
- ✓ Ability to have a technical conversation about ML, LLMs, RAG, and MLOps
- ✓ A LinkedIn post / case study showing your 30-day journey — strong signal for hiring managers

What comes after day 30? ↗

